UZMAN ALI

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Profile Summary: RF and Antenna Engineer with 6+ years of experience in antenna design, RF development, electromagnetic simulation, prototyping, laboratory validation, and scientific research. Experienced in HFSS, CST, RF measurements, microwave systems, implantable antennas, wireless communication, SAR analysis, and experimental verification. Strong background in translating scientific research into practical RF solutions through collaboration with academia and industry.

Technical Expertise: • Antenna Design • RF Systems • Microwave Engineering • Electromagnetic Simulations (HFSS/CST/Sim4Life) • RF and Microwave Circuit Design • RF Measurements • Anechoic Chamber • Vector Analyzer • MATLAB • SAR Analysis •Prototype Development • Experimental Validation • Presentation Skills • Scientific Reports • Research Project Development • Industrial Collaboration

EEDUCATION

- 27 September 2026 (Thesis submitted), Doctor of Science (Technology), Computing and Electrical Engineering with emphasis on Antennas and Bio-electromagnetics, Tampere University
- 30 October 2019, Master of Science (Technology), Electrical Engineering with majors in Wireless Communication and RF Systems, Tampere University

WORK EXPERIENCE

April 2023-Aug 2026, Full-Time Doctoral Researcher, Tampere University

- Conducted research on RF systems, antennas, electromagnetic propagation, and wireless communications, combining simulation, prototyping, and experimental validation
- Designed, optimized, and characterized multiband and broadband antenna systems using ANSYS HFSS and electromagnetic modelling techniques
- Developed and implemented RF measurement methodologies and experimental setups for wireless communication and antenna characterization
- Performed RF testing, VNA measurements, data acquisition, and result analysis for prototype validation
- Developed MATLAB tools for processing, visualization, and analysis of large electromagnetic and measurement datasets
- Published and presented research results in leading IEEE journals and international conferences including EuCAP, APS/URSI, and APMC
- Supervised students and research trainees in antenna design, EM simulation, RF measurements, and scientific publishing

January 2022-April 2023, Full-Time Application Engineer (High-Frequency), EDR & Medeso Oy

- Supported industrial customers in RF, microwave, antenna, and electromagnetic simulation projects using ANSYS HFSS and multiphysics simulation tools
- Provided technical consulting, troubleshooting, and workflow optimization for high-frequency simulation and product development activities
- Supported antenna design, RF system analysis, and electromagnetic compatibility assessments through simulation-driven engineering approaches
- Conducted technical demonstrations, customer training sessions, and workshops related to RF and electromagnetic simulation technologies
- Collaborated with industrial R&D teams to accelerate design validation and product development processes

April 2020- July 2021, Full-Time Antenna Design Engineer, Radientum Oy

- Designed and optimized antennas for commercial wireless and IoT products using full-wave electromagnetic simulations and experimental validation
- Performed antenna matching, tuning, impedance optimization, and performance evaluation across multiple frequency bands
- Conducted antenna characterization and RF measurements to verify compliance with project specifications and performance requirements
- Supported prototype development, testing, and integration activities in multidisciplinary product development projects
- Contributed to the development of practical antenna solutions for wireless communications and connected devices

May 2019-October 2019, Research Assistant, Tampere University

- Antenna designing, RF measurements, electromagnetic simulations, and laboratory testing activities
- Supported wireless communication research projects through data analysis, experimental validation, and technical reporting

CAREER BREAKS

Laid-off during COVID-19 pandemic at Radientum. However, this period strengthened my self-management, resilience, and ability to remain focused in high pressure situations.

SCHOLARSHIPS AND GRANTS

Scholarships:

- Awarded a 100% tuition fee waiver for master's studies at Tampere University

Grants:

- €1,000 DELTA doctoral training grant for participation in the European school of Antennas and Propagation "Body-Centric Wireless Communications" at Queen Mary University of London
- Awarded a competitive €26,000 grant by Kaute-säätiö for independent research aimed at addressing key research gaps identified through my PhD work

RESEARCH OUTPUT

Publications:

- Author and co-author of multiple peer-reviewed journal articles in IEEE AWPL, IEEE J-ERM, IEEE Open-Acess and conference papers in leading IEEE venues on antenna design, RF systems, wireless communications, electromagnetic safety, and experimental validation
- Presented research findings at major international conferences, including EuCAP, APS/URSI, and APMC
- A detailed publication list is attached as a separate document

Creative Work:
Engineered a multifunctional antenna measurement setup with independent control of antenna height, spacing, axial rotation, and planar orientation to enable accurate wireless communication experiments.

RESEARCH SUPERVISION AND COLLABORARTION

- Supervised and mentored master's students in antenna simulations, measurements, thesis writing, presentations, and scientific publishing

- Collaborated with exchange doctoral researchers on RF energy harvesting, implantable MIMO antennas, and wireless power transfer
- International research partnerships leading to peer-reviewed publications in antenna technologies
- Fostered collaboration and knowledge exchange with bachelor's students from Pakistan, supporting their development of research skills and transition into academic research

TEACHING MERITS

Teaching:

- Assisted in teaching the course RF Circuit Design, supporting student learning exercise and practical sessions, including guidance on their semester projects and reports (Autumn 2024)
- Teacher assistant to the course of Basics of RF and Antenna Engineering and conducting computer exercise sessions including HFSS and ADS simulation use (Autumn 2026)
- Conducted hands-on demonstrations of RF measurement equipment, including Vector Network Analyzers, and provided instruction in soldering and prototyping techniques (Autumn 2025)

Tutoring:

- Mentoring and supporting bachelor's students from diverse backgrounds, acting as a liaison between students and faculty, and provided guidance for the start of their studies
- Assisted new international master's students at Tampere University with campus facilities, onboarding procedures, and adapting to student life in Finland

SCIENTIFIC ACTIVITIES

- Presented research findings (oral and poster presentations) at international conferences, including EuCAP 2024, and 2025, APMC 2024, and 2025, and APS-URSI 2024
- Presented our research progress in national and institutional seminars, such as RF Sampo 2024 (Oulu), Research Day 2024,2025 (Tampere University) and LFTY 2024 (Tampere)
- Participated and presented in doctoral student seminars, including DELTA Autumn School 2023 (Jyväskylä), DELTA Summer School 2024 (Tampere), 2025 (Turku), 2026 (Helsinki)
- Attended industrial and academic workshops, including Ansys Learning (Tampere 2024-2026), Simulia Workshop (Oulu 2024), and several workshops during EuCAP 2024,2025, APS/URSI 2024, and APMC 2025
- Participated in summer schools for advanced antenna research, including the European School of Antennas and Propagation "Body-Centric Wireless Communications" at Queen Mary University of London and DELTA Winter School 2024 and 2026 in Ruka

Peer Review Activities:

Served as a reviewer for over 20 conference and journal manuscripts, including contributions to IEEE AWPL, IEEE ISBI, IEEE J-ERM, and Communication Engineering

Memberships:

IEEE, IEEE Antennas and Propagation Society Membership, IEEE Engineering Medicine and Biology Society, IEEE Microwave Theory and Technology Society Membership, IEEE Young Professionals

REFERENHCES

PhD Supervisor

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